from its location to end-users, which are often in other locations. The new decentralized consolidation model will speed up the dissemination of consolidated market data by allowing competing consolidators to collect data directly from each SRO and consolidate the data in the same data center as end users. Latency-sensitive data end-users will be able to receive consolidated market data products at the same data center location from which the competing consolidator operates.

Under this new model, the relevant exchange will provide quotes and trades in the NMS stocks they trade directly to competing consolidators and self-aggregators and the hub-and-spoke method of centralized collection and dissemination will be eliminated.[543] Further, by fostering a competitive environment for the collection, consolidation, and dissemination of consolidated market data, the decentralized consolidation model will incentivize greater innovation, competitive pricing, and the timely adoption of updated technologies into the national market system.

The Commission believes that the decentralized consolidation model will better serve the needs of market participants and investors. It should address concerns about the costs associated with the current structure, in which many market participants are compelled to buy proprietary feeds and the exclusive SIP feeds to trade competitively and represent their customers' orders.[544]

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543 See Proposing Release, 85 FR at n.395. Under the centralized consolidation model, quotes and trades that occur on Nasdaq for NYSE-listed stocks must be provided to the CTA/CQ SIP for dissemination. Under the decentralized consolidation model, such quotes and trades in NYSE-listed stocks will be provided directly by Nasdaq to competing consolidators and self-aggregators.

544 One commenter stated that it expects that its use of direct feeds would be eliminated if the proposal is implemented. See NBIM Letter at 4.

The amendments also should address the concerns about, and improve, the content and latency differentials that currently exist between SIP data and proprietary data.[545] The Commission believes that the amendments will provide all market participants with access to a real-time stream of consolidated market data, improve the national market system, help to ensure the continued success of the U.S. securities markets, and better achieve the goals of Section 11A of the Exchange Act by assuring "the availability to brokers, dealers and investors of information with respect to quotations for and transactions in securities" that is prompt, accurate, reliable, and fair.

B. Proposed Decentralized Consolidation Model

The Commission proposed a decentralized consolidation model in which new competing SIPs, called competing consolidators, would collect the data content underlying consolidated market data from the individual SROs, consolidate the information of all of the SROs, and disseminate that consolidated information as consolidated market data to end users. The proposed decentralized consolidation model also would allow broker-dealers to act as self-aggregators to collect all of the data content underlying consolidated market data from the individual SROs and consolidate that information solely for their internal use. The Commission proposed this model to reduce significantly the geographic and other latencies inherent in the existing centralized consolidation model. The proposed decentralized consolidation model would allow competing consolidators and self-aggregators to eliminate the back-and-forth travel of data associated with the centralized consolidation model, to operate in the data center of their

545 See Proposing Release, 85 FR at 16764–65.

choice (i.e., in close proximity to data subscribers), and to foster a competitive environment for the aggregation and transmission of consolidated market data.[546]

1. Comments on the Decentralized Consolidation Model

The Commission received comments on the proposed decentralized consolidation model.[547] Many commenters supported the goals of the decentralized consolidation model and the potential positive impacts this model would have on the provision of consolidated market

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546 See Proposing Release, 85 FR at n.419–20.

547 See BlackRock Letter; Fidelity Letter; State Street Letter; Wellington Letter; ICI Letter; Virtu Letter; AHSAT Letter; IntelligentCross Letter; BestEx Research Letter; MEMX Letter; Clearpool Letter; T. Rowe Price Letter; ACS Execution Services Letter; IEX Letter; SIFMA Letter; MFA Letter; Schwab Letter; RBC Letter; STANY Letter II; Angel Letter; Nasdaq Letter III; Nasdaq Letter IV; NYSE Letter II; FINRA Letter; Cboe Letter; Proof Trading Letter; Citadel Letter; TD Ameritrade Letter; Data Boiler Letter I; Healthy Markets Letter I; Joint CRO Letter; Susquehanna Letter; NovaSparks Letter; Better Markets Letter; Capital Group Letter; McKay Letter; NBIM Letter; Wharton Letter; letter from Kermit R. Kubitz, dated May 26, 2020, ("Kubitz Letter"); letter from Kelvin To, Founder and President, Data Boiler Technologies, LLC, dated June 10, 2020, ("Data Boiler Letter II"); DOJ Letter; letter from Nandini Sukumar, Chief Executive Officer, World Federation of Exchanges, to Chairman Clayton and Commissioners Lee, Peirce, and Roisman, dated May 26, 2020, ("WFE Letter"); letters to Vanessa Countryman, Secretary, Commission, from Stephen J. McNeany Chief Executive Officer, and Frank W. Piasecki, President, ACTIV Financial Systems, Inc., dated May 26, 2020, ("ACTIV Financial Letter"); Doris Choi, Co-General Counsel, ICE Data Services, dated May 29, 2020, ("IDS Letter I"); John L. Thornton and R. Glenn Hubbard, Co-Chairs, and Hal S. Scott, President, Committee on Capital Markets Regulation, dated Apr. 23, 2020, ("Committee on Capital Markets Regulation Letter"); Kevin R. Edgar, Counsel, BakerHostetler LLP and Counsel, Equity Markets Association, dated June 30, 2020, ("Equity Markets Association Letter"); Joanna Mallers, Secretary, FIA Principal Traders Group, dated June 3, 2020, ("FIA PTG Letter"); Tom C. W. Lin, Professor of Law, Temple University Beasley School of Law, dated May 26, 2020, ("Temple University Letter"); Tyler Gellasch, Executive Director, Healthy Markets, dated July 27, 2020, ("Healthy Markets Letter II"); Doris Choi, Co-General Counsel, ICE Data Services, dated Aug. 12, 2020, ("IDS Letter II").

data,[548] while numerous commenters raised issues with—or questioned certain aspects of—the proposed model.[549]

Commenters that supported the decentralized consolidation model believed that it would inject needed competition into the consolidated market data environment,[550] address conflicts of interest in the centralized consolidation model,[551] reduce latency in the dissemination of consolidated market data,[552] improve the usefulness of consolidated market data as an alternative to proprietary market data feeds,[553] improve the reliability of the consolidated market data

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548 See BlackRock Letter; Fidelity Letter; State Street Letter; Wellington Letter; ICI Letter; Virtu Letter; AHSAT Letter; FIA PTG Letter; IntelligentCross Letter; Committee on Capital Markets Regulation Letter; BestEx Research Letter; Wharton Letter; MEMX Letter; Clearpool Letter; T. Rowe Price Letter; Capital Group Letter; DOJ Letter; ACS Execution Services Letter; IEX Letter; SIFMA Letter; ACTIV Financial Letter; MFA Letter; Better Markets Letter; NBIM Letter; NovaSparks Letter; letter from Anthony H Steinmetz, dated Feb. 17, 2020, ("Steinmetz Letter") (supporting the proposal generally).

549 See STANY Letter II; Angel Letter; Nasdaq Letter III; Nasdaq Letter IV; NYSE Letter II; FINRA Letter; Cboe Letter; Proof Trading Letter; Citadel Letter; TD Ameritrade Letter; Kubitz Letter; Data Boiler Letter I; Data Boiler Letter II; Healthy Markets Letter I; WFE Letter; Joint CRO Letter; Equity Markets Association Letter.

550 See MEMX Letter at 3, 8; Committee on Capital Markets Regulation Letter at 3; BestEx Research Letter at 1; State Street Letter at 3; ACTIV Financial Letter at 1; Fidelity Letter at 9; SIFMA Letter at 5, 12; Wellington Letter at 1; IntelligentCross Letter at 4–5; ICI Letter at 4, 10; RBC Letter at 6; DOJ Letter at 5; Capital Group Letter at 4; T. Rowe Price Letter at 4; Virtu Letter at 6.

551 See SIFMA Letter at 5; Fidelity Letter at 3, 10; IEX Letter at 2.

552 See MEMX Letter at 6, 7, 8; Fidelity Letter at 3, 10; Wellington Letter at 1; ICI Letter at 4, 10; Capital Group Letter at 4; BlackRock Letter at 5; IEX Letter at 3; Better Markets Letter at 3; AHSAT Letter at 1, 3; DOJ Letter at 3, 4; SIFMA Letter at 1, 5, 11; ACS Execution Services Letter at 5.

553 See MFA Letter at 2; Capital Group Letter at 2, 4; ICI Letter at 4; DOJ Letter at 2–3, 4; SIFMA Letter at 5; MEMX Letter at 2, 8; NBIM Letter at 4.

infrastructure,[554] and reduce the cost of consolidated market data[555] while increasing its quality.[556]

Commenters that raised concerns about the proposed decentralized consolidation model said that it would not achieve its goal of disseminating consolidated market data to market participants in a more timely, efficient, and cost-effective manner than the current centralized consolidation model;[557] that its impact on the markets would be uncertain until implementation;[558] that the impact on fees and costs for consolidated market data was uncertain;[559] that the new architecture could result in increased costs for some market participants;[560] and that it could result in increased costs for SROs.[561] Commenters also

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554 See NovaSparks Letter at 1.

555 See BestEx Research Letter at 1, 4; DOJ Letter at 3–4, 5; Committee on Capital Markets Letter at 3; IntelligentCross Letter at 5; Better Markets Letter at 3; RBC Letter at 5–6; State Street Letter at 3; Fidelity Letter at 3, 9; Wellington Letter at 1; BlackRock Letter at 5; IEX Letter at 3; SIG Letter at 1.

556 See Committee on Capital Markets Regulation Letter at 3; BestEx Research Letter at 1; RBC Letter at 5–6; State Street Letter at 3.

557 See Healthy Markets Letter I at 2; Kubitz Letter at 1; Data Boiler Letter I at 46–47; Data Boiler Letter II at 1; Citadel Letter at 5; TD Ameritrade Letter at 2; NYSE Letter II at 22; Nasdaq Letter IV at 2–3, 8; Angel Letter at 18–20; STANY Letter II at 5.

558 See FINRA Letter at 3, 4; letter from Linda Moore, President and CEO, TechNet, to Vanessa Countryman, Secretary, Commission, dated June 18, 2020, ("TechNet Letter II") at 2.

559 See STANY Letter II at 5; Data Boiler Letter I at 47; TD Ameritrade Letter at 15; Nasdaq Letter IV at 23, 26, 47–48, 60. See also, e.g., Angel Letter at 22–23.

560 See Cboe Letter at 23–24; FINRA Letter at 3, 4. See also, e.g., Angel Letter at 21.

561 See FINRA Letter at 3, 4; Nasdaq Letter IV at 27, 29, 30.

questioned its feasibility,[562] the complexity introduced by multiple competing consolidators,[563] its impact on regulation,[564] and its benefits to latency.[565] A few commenters also suggested that the decentralized consolidation model would benefit from further consideration by market participants and the Commission and should be the subject of a separate rulemaking.[566]

These comments are addressed below.

2. Comments on the Effectiveness of the Proposal

Several commenters questioned whether the proposed decentralized consolidation model would achieve its goal of disseminating consolidated market data to market participants in a more timely, efficient, and cost-effective manner than the current centralized consolidation model.[567] One commenter stated that while the proposed system may be desirable, "it is not clear how the development of multiple parties interacting and providing quotations and trade

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562 See Nasdaq Letter IV at 23–26; NYSE Letter II at 13, 17–18; IDS Letter I at 3, 4, 7–8; IDS Letter II at 1, 3; STANY Letter II at 6; Data Boiler Letter I at 46; Angel Letter at 18, 20; Equity Markets Association Letter at 3; FINRA Letter at 3; TechNet Letter II at 1–2.

563 See FINRA Letter at 2, 3, 5–6; Angel Letter at 18–19; Healthy Markets Letter I at 4–5; TechNet Letter II at 2; STANY Letter II at 6, 8; Joint CRO Letter at 2; Data Boiler Letter I at 48; Nasdaq Letter III at 8; Citadel Letter at 5; TD Ameritrade Letter at 12–13; WFE Letter at 1.

564 See Nasdaq Letter IV at 2, 3, 4, 12–13, 35; Joint CRO Letter at 2, 3, 4; FINRA Letter at 4, 5, 6; TechNet Letter II at 2; Data Boiler Letter I at 48; Healthy Markets Letter I at 4–5; TD Ameritrade Letter at 13; Citadel Letter at 5; Kubitz Letter at 1; NYSE Letter II at 23.

565 See Cboe Letter at 23; Nasdaq Letter IV at 49; STANY Letter II at 5, 6; Citadel Letter at 5; NYSE Letter II at 11, 22, 23; TD Ameritrade Letter at 12; Proof Trading Letter at 1; Angel Letter at 18, 19; FINRA Letter at 8; IDS Letter I at 15; Data Boiler Letter II at 1.

566 See Citadel Letter at 5; STANY Letter II at 8.

567 See Healthy Markets Letter I at 2; Kubitz Letter at 1; Data Boiler Letter I at 46–47; Data Boiler Letter II at 1; Citadel Letter at 5; TD Ameritrade Letter at 2, 12; NYSE Letter II at 22; Nasdaq Letter IV at 2–3, 8; Angel Letter at 18–20; STANY Letter II at 5.

data can be implemented over time to assure accuracy, completeness and avoidance of gaming and fraud."[568] Another commenter said that the proposed decentralized consolidation model is impractical and would increase market fragmentation.[569] One commenter stated that the Commission failed to show how the decentralized consolidation model would result in the dissemination of market data that is prompt, accurate, reliable, and fair and said that the proposal violates the Administrative Procedure Act ("APA")[570] because it did not demonstrate a rational connection between the proposal and its goals.[571]

A few commenters stated that the Proposing Release did not include enough information to evaluate whether the decentralized consolidation model would reduce market data costs, improve transmission latency, and improve resiliency.[572] One commenter stated that the proposal lacked information about "data quality, availability, reliability and potential for significant additional cost" and expressed the view that the negative effects of the proposal could exceed any benefits to retail investors.[573] Another commenter said that it was "unclear whether the prices set by competing consolidators will be reliable, resilient, or well-regulated; and how anomalies and disparities among competing consolidators will be resolved."[574]

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568 Kubitz Letter at 1.

569 See Data Boiler Letter II at 1.

570 5 U.S.C. 551 et seq.

571 See NYSE Letter II at 22.

572 See STANY Letter II at 5; TD Ameritrade Letter at 2.

573 TD Ameritrade Letter at 2.

574 TechNet Letter II at 2.

Some commenters raised questions regarding the competitive aspects of the proposal. One commenter stated that there was no guarantee that competition would improve latency and cost,[575] and another commenter questioned the ability of competing consolidators to provide the needed competition to decrease latency and cost.[576] Further, one commenter stated that, although the proposal would provide for competition, there is no guarantee that competition would occur or that the proposal would result in a competitive outcome that would benefit investors.[577] This commenter stated that competition would result in competing consolidators selling differentiated products that would result in a proliferation of market data tiers and information asymmetries.[578]

However, several commenters said that the proposed decentralized consolidation model would introduce needed competition,[579] which would result in better quality consolidated market

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575 See TD Ameritrade Letter at 12.

576 See Data Boiler Letter I at 46–47.

577 See Nasdaq Letter IV at 3.

578 See id.

579 See MEMX Letter at 3, 8; T. Rowe Price Letter at 4; Committee on Capital Markets Regulation Letter at 3; BestEx Research Letter at 1; State Street Letter at 3; ACTIV Financial Letter at 1; Fidelity Letter at 9; SIFMA Letter at 5, 12; Wellington Letter at 1; IntelligentCross Letter at 4–5; ICI Letter at 4, 10; RBC Letter at 6; DOJ Letter at 5; Capital Group Letter at 4; Virtu Letter at 6.

data,[580] lower market data costs,[581] and improved latency.[582] One commenter stated that competition in the consolidation and dissemination of market data would increase investor choice and would address both the conflicts of interest that exist in the centralized consolidation model and the latency advantages enjoyed by market participants that are able to purchase proprietary data feeds.[583] One commenter stated that competition will allow for innovation that could reduce dependence on proprietary market data,[584] and another asserted that "a market with competing data feeds will be more efficient and effective."[585] One commenter said that modernization through competitive market forces would bring desired changes to consolidated

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580 See Committee on Capital Markets Regulation Letter at 3; BestEx Research Letter at 1; RBC Letter at 6; State Street Letter at 3.

581 See Fidelity Letter at 3, 9, 10; Committee on Capital Markets Regulation Letter at 3; BestEx Research Letter at 1, 4; ACTIV Financial Letter at 1; SIFMA Letter at 12; State Street Letter at 3; Wellington Letter at 1; IntelligentCross Letter at 5; ICI Letter at 4; RBC Letter at 5–6; DOJ Letter at 3–4, 5; Better Markets Letter at 3; BlackRock Letter at 5; IEX Letter at 3.

582 See SIFMA Letter at 1, 5, 11; ICI Letter at 4, 10; Capital Group Letter at 4; MEMX Letter at 8; Fidelity Letter at 3, 10; BlackRock Letter at 5; IEX Letter at 3; Better Markets Letter at 3; AHSAT Letter at 1; DOJ Letter at 3–4, Wellington Letter at 1 ("We believe the introduction of competitive forces to the distribution of data will result in lower-latency, faster data that is more broadly available and also at reduced costs for participants."); NovaSparks Letter at 1 ("The competitive nature of the new model will encourage Competing Consolidators to deliver excellent reliability, functionality and performance.").

583 See SIFMA Letter at 5, 12.

584 See State Street Letter at 3.

585 Capital Group Letter at 4.

market data and its framework,[586] and another supported the proposal's addition of competition "while still preserving a significant role for the exchanges to participate."[587]

Several commenters stated that the proposed decentralized consolidation model could improve the usefulness of consolidated market data and make it a viable alternative to proprietary market data feeds.[588] Commenters indicated that the exchange operators of the exclusive SIPs currently lack the incentive to improve the content, delivery, and pricing of consolidated market data because improved consolidated market data could reduce demand for the proprietary data feeds that the exchanges sell.[589] Two commenters stated that the proposal would result in a less costly alternative to proprietary market data feeds.[590] One of these commenters stated that alternatives to proprietary data feeds could increase participation in the financial services industry and bring down costs for market participants.[591]

Two commenters stated that the proposal would narrow the content and latency gaps between proprietary market data feeds and consolidated market data.[592] One commenter said

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586 See T. Rowe Price Letter at 4.

587 Virtu Letter at 6.

588 See MFA Letter at 2; Capital Group Letter at 2, 4; ICI Letter at 4; DOJ Letter at 2–3, 4; SIFMA Letter at 5; MEMX Letter at 2, 8; NBIM Letter at 4, 5–6.

589 See BestEx Research Letter at 1; State Street Letter at 3.

590 See DOJ Letter at 4; MEMX Letter at 8 ("The new content and infrastructure enhancements would provide an opportunity to introduce new less-expensive NMS data alternatives to proprietary market data products.").

591 See DOJ Letter at 4.

592 See MFA Letter at 2 (stating that the proposal should narrow the "significant gap in usefulness between exchange proprietary data feeds and consolidated market data"); Capital Group Letter at 2.